Turbocharged and tiny

Ubiquitous, that's what they are. Open your eyes and you'll spot one, you probably have at least one on your person or around you at any given time. It may not be immediately apparent where and how they play a significant role. Yep we're talking about the all-powerful SoC or System On Chip to the uninitiated.

You probably mistook SoCs to be just processors. A processor or micro-processor is just one part of the average SoC. There's a lot more squeezed into an SoC's tiny stamp sized area and it requires quite a fair amount of digging to completely fathom the magnitude of what's happening at that microscopic level.

That's where we come into the picture.

We start off with an in-depth explanation on what SoCs are and how there are very similar circuits that often get mistaken for each other. There are SiPs, SoPs, MCMs and many more circuits which all come in similarly small packages and do practically the same thing, however, we delve into the nuances of what makes each of them stand apart from one another. Then we move on to a history lesson into how SoCs came to be, what necessitated their formation and how they've grown over the years. We then get into the nitty gritties of each hardware component that goes into making an SoC. This is where you'll learn how an SoC is essentially an entire computer sans the display, keyboard and mouse. All that power crammed into a really small space.

What's hardware supposed to do without the accompanying software? When we use the term software, we really mean the whole shebang – from the OS Kernel to the drivers to the runtimes to the APIs, to the frameworks and finally culminating in the application itself. Each of these play their important role in the entire process.